Task 5 – Fingerprint (Arabic Dialects Detection)

The Arabic language is rich of its spoken variations, i.e. dialects, from the ocean to the gulf. Each dialect has its intrinsic features that can be used to identify a dialect from others. Develop a web application that perform the following tasks:

1. Open a voice file, around 30sec of spoken Arabic, and visualize its spectrogram. You should be ready with 4 different dialects, each spoken by 4 different voices.
2. Use classic machine learning, i.e. NO AI models or deep learning networks, to identify the file spoken dialect. Visualize the governing features that led to your decision either on the signal or the spectrogram or whatever features you chose. Your submission will not be accepted if the displayed features do not clearly show distinguishable differences between dialects.
3. Display the sentences/words that are pronounced from the opened file in real time while it is pronounced. You can use any external pretrained model or engines for this purpose. You just need to connect to that helper via solid APIs.
4. Allow the user to choose another dialect that s/he wishes to hear the same file with. This includes changing the tones as well as the words (each dialect has different words dictionary). Please, be ready with examples that illustrates all possibilities.
5. Your program should be able to take two files, make a weighted average of them (there should be a slider to control the weighting percentage). Then, your software should treat the new summation as a new file and search for the closest dialects to it. One would expect the classifier output to be a mix of the dialects of the contributing files based on the weighting percentage of each.